

Choosing the beamformer settings

A plain-language summary of the test. Research and educational use only, not a medical diagnosis.

1. The short answer

The three imaging methods (DAS, DMAS and DMAS-D4) each have a few number settings, or "knobs". Every value from 0 to 1 was tried, in steps of 0.05, on real breast-phantom scans. The value that worked best **on average across all scans** was kept.

Knob	Textbook value	Chosen value	In one line
DAS dimming weight	0	0.55	Some dimming helps; too much flags healthy scans
DMAS pair squeeze	0.50	0.30	Squeezing a bit more put the spot closer to the tumor
DMAS dimming weight	0	0.00	Extra dimming did not help DMAS
D4 compression	0.25	0.25	The textbook value was already the best

Overall score out of 100 (higher is better) for each full set of settings:

Settings	DAS score	DMAS + D4 score
Textbook	56.6	59.1
Chosen	64.4	61.8

2. How the test worked

Scans used: 12 scans from the UM-BMID phantom dataset (9 with a tumor, 3 healthy), taken from all three dataset generations and covering small, medium and large tumors in different positions.

For each knob value: the picture was rebuilt for every scan, and the program found the brightest spot **on its own**, without being told where the tumor really was. Only afterwards was the answer checked:

What was checked	Good result
Tumor scans: how far the bright spot landed from the real tumor (cm)	Small distance
Tumor scans: whether the spot is small and clear, not smeared or buried in edge noise	Small, sharp spot
Healthy scans: whether the program wrongly said "tumor"	No false alarm

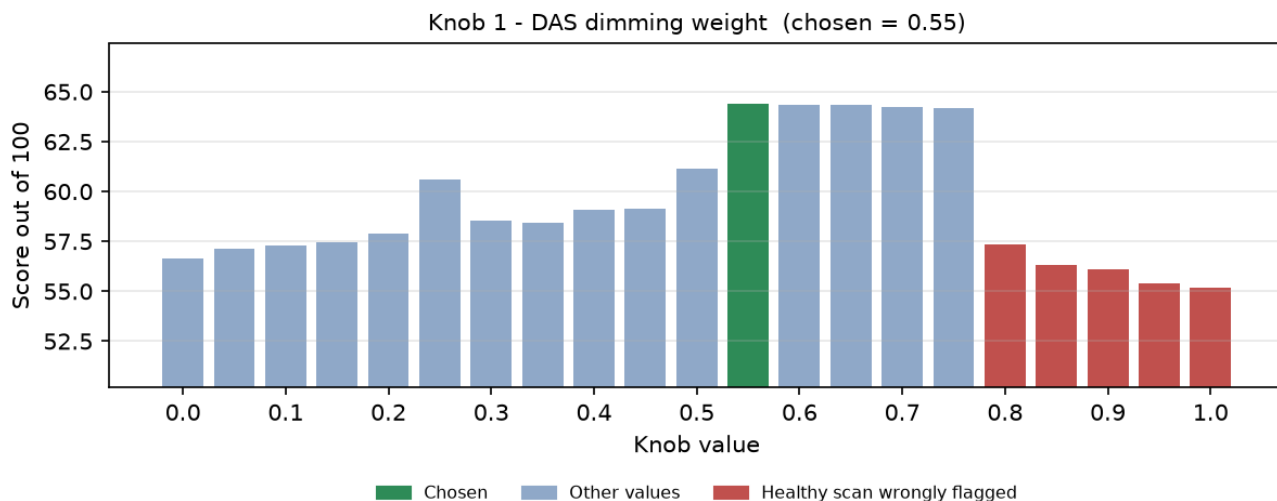
These were combined into one **score out of 100**. Tumor scans count for 70% and healthy scans for 30%. The chosen value is the one with the **highest average score over all scans**, not the best value for any single scan.

3. Each knob, and why the other values lost

Each chart shows the score for all 21 values. **Green** is the chosen value. **Red** means at least one healthy scan was wrongly flagged as a tumor. The table below each chart lists the most discussed values. "Average miss" is how far, on average, the bright spot landed from the real tumor.

Knob 1 - DAS dimming weight

DAS adds up the echoes from all antennas. This knob decides how much to dim a spot when the antennas disagree about it. 0 means no dimming (the textbook version). 1 means strong dimming.



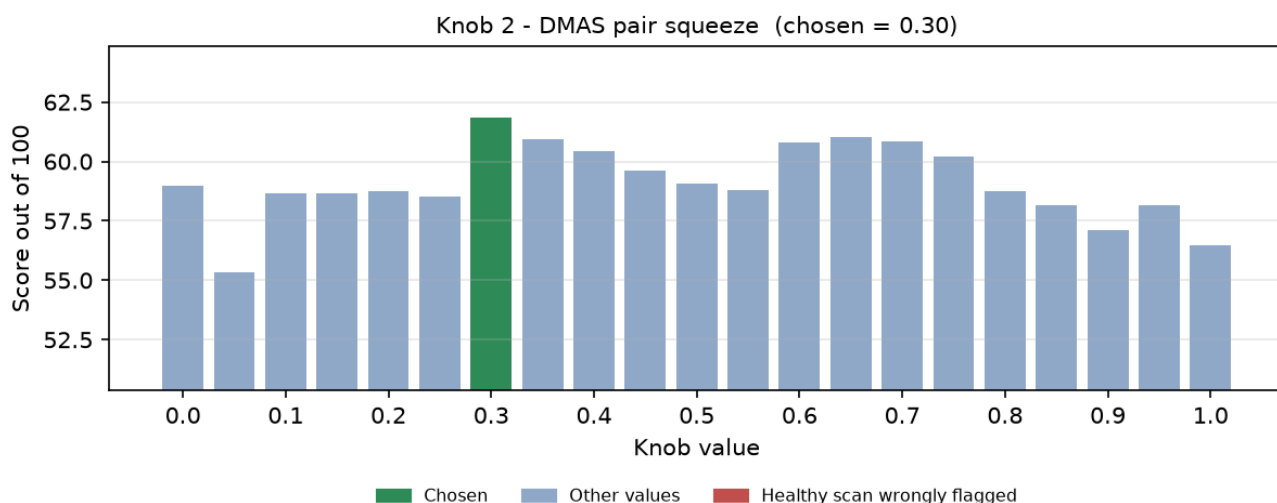
Value	Score /100	Average miss	Healthy false alarms	What happened
0.00 (textbook)	56.6	5.3 cm	0 of 3	Bright spot lands farther from the tumor
0.25	60.6	4.2 cm	0 of 3	Bright spot lands farther from the tumor
0.35	58.5	4.5 cm	0 of 3	Bright spot lands farther from the tumor
0.50	61.1	3.9 cm	0 of 3	Bright spot lands farther from the tumor
0.55 (chosen)	64.4	3.4 cm	0 of 3	Best average score
0.75	64.2	3.3 cm	0 of 3	Almost the same score - difference too small to matter
1.00	55.2	3.7 cm	1 of 3	A healthy scan is wrongly flagged as tumor

Why not the other values?

- **0.00 to 0.50**: the bright spot landed farther from the real tumor, about 4.7 cm away instead of 3.4 cm.
- **0.60 to 0.75**: almost the same score (64.3 vs 64.4). The gap is too small to call a real difference, so the top value was kept.
- **0.80 to 1.00**: a healthy scan was wrongly called a tumor (1 of 3 healthy scans).

Knob 2 - DMAS pair squeeze

DMAS multiplies the echoes of every pair of antennas. This knob sets how much each pair's result is squeezed before everything is added up. 0.5 is the textbook square root. A smaller number squeezes more, a larger number squeezes less.



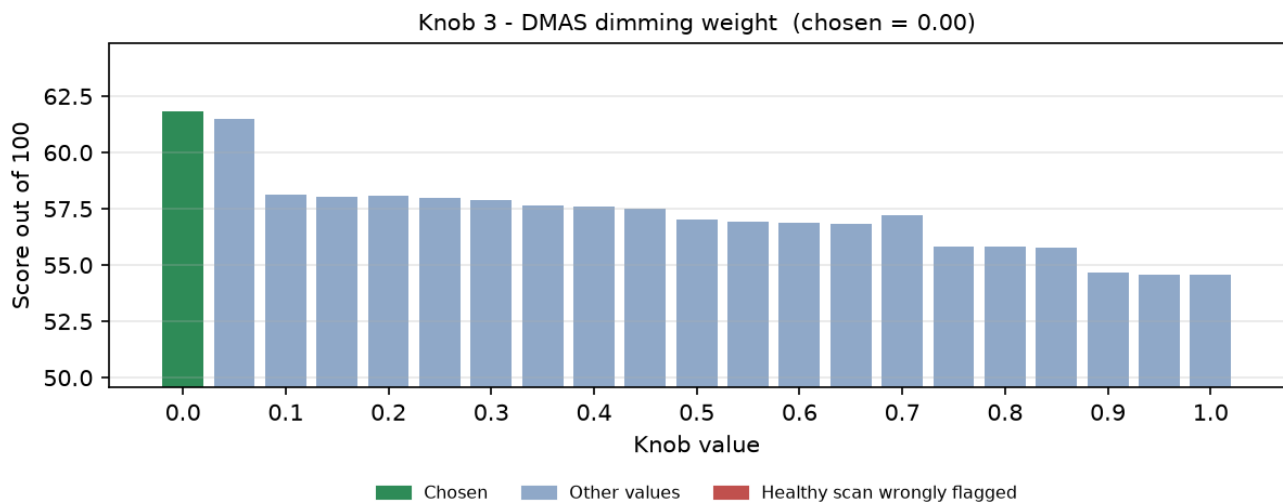
Value	Score /100	Average miss	Healthy false alarms	What happened
0.00	59.0	4.5 cm	0 of 3	Bright spot lands farther from the tumor
0.25	58.5	4.3 cm	0 of 3	Bright spot lands farther from the tumor
0.30 (chosen)	61.8	3.2 cm	0 of 3	Best average score
0.35	61.0	3.4 cm	0 of 3	Almost the same score - difference too small to matter
0.50 (textbook)	59.1	3.6 cm	0 of 3	Slightly lower score, no single big problem
0.55	58.8	3.7 cm	0 of 3	Slightly lower score, no single big problem
0.75	60.2	3.3 cm	0 of 3	Slightly lower score, no single big problem
1.00	56.5	3.9 cm	0 of 3	Bright spot lands farther from the tumor

Why not the other values?

- **0.00 to 0.25:** the bright spot landed farther from the real tumor, about 4.4 cm away instead of 3.2 cm.
- **0.35:** almost the same score (61.0 vs 61.8). The gap is too small to call a real difference, so the top value was kept.
- **0.40 to 0.60:** slightly lower score overall (59.7 vs 61.8), with no single big problem.
- **0.65 to 0.70:** almost the same score (60.9 vs 61.8). The gap is too small to call a real difference, so the top value was kept.
- **0.75 to 0.95:** slightly lower score overall (58.5 vs 61.8), with no single big problem.
- **1.00:** the bright spot landed farther from the real tumor, about 3.9 cm away instead of 3.2 cm.

Knob 3 - DMAS dimming weight

The same dimming idea as Knob 1, but applied to the DMAS picture. 0 means no dimming (textbook). 1 means strong dimming.



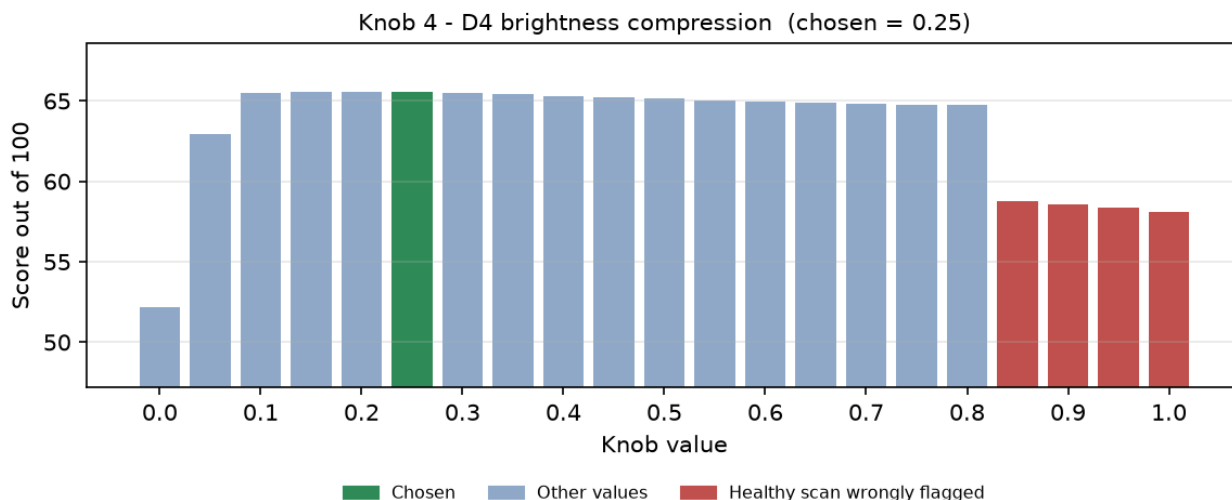
Value	Score /100	Average miss	Healthy false alarms	What happened
0.00 (textbook, chosen)	61.8	3.2 cm	0 of 3	Best average score
0.25	58.0	4.1 cm	0 of 3	Bright spot lands farther from the tumor
0.35	57.7	4.2 cm	0 of 3	Bright spot lands farther from the tumor
0.50	57.0	4.2 cm	0 of 3	Bright spot lands farther from the tumor
0.60	56.9	4.2 cm	0 of 3	Bright spot lands farther from the tumor
0.75	55.8	4.1 cm	0 of 3	Bright spot lands farther from the tumor
1.00	54.6	4.2 cm	0 of 3	Bright spot lands farther from the tumor

Why not the other values?

- **0.05:** almost the same score (61.5 vs 61.8). The gap is too small to call a real difference, so the top value was kept.
- **0.10 to 1.00:** the bright spot landed farther from the real tumor, about 4.2 cm away instead of 3.2 cm.

Knob 4 - D4 brightness compression

After DMAS, D4 compresses the brightness range of the final picture so weak and strong spots are easier to compare. 0.25 is the textbook fourth root. 1 means no compression at all.



Value	Score /100	Average miss	Healthy false alarms	What happened
0.00	52.2	9.4 cm	0 of 3	Bright spot lands farther from the tumor
0.25 (textbook, chosen)	65.5	3.2 cm	0 of 3	Best average score
0.35	65.4	3.2 cm	0 of 3	Almost the same score - difference too small to matter
0.50	65.1	3.2 cm	0 of 3	Almost the same score - difference too small to matter
0.55	65.0	3.2 cm	0 of 3	Almost the same score - difference too small to matter
0.75	64.8	3.2 cm	0 of 3	Almost the same score - difference too small to matter
1.00	58.1	3.2 cm	1 of 3	A healthy scan is wrongly flagged as tumor

Why not the other values?

- **0.00**: the bright spot landed farther from the real tumor, about 9.4 cm away instead of 3.2 cm.
- **0.05**: slightly lower score overall (63.0 vs 65.5), with no single big problem.
- **0.10 to 0.20**: almost the same score (65.5 vs 65.5). The gap is too small to call a real difference, so the top value was kept.
- **0.30 to 0.80**: almost the same score (65.1 vs 65.5). The gap is too small to call a real difference, so the top value was kept.
- **0.85 to 1.00**: a healthy scan was wrongly called a tumor (1 of 3 healthy scans).

4. Scan by scan: textbook settings vs chosen settings

Distance (cm) from the program's bright spot to the real tumor, for each tumor scan. Smaller is better. The last column is the DAS dimming weight that would suit only that one scan.

Scan	Tumor	DAS textbook	DAS chosen	D4 textbook	D4 chosen	Best DAS weight for this scan alone
gen1 #3	2 cm, lower-right	1.7	2.3	2.2	2.2	0.15
gen1 #7	4 cm, upper-left	9.8	6.7	6.7	6.7	0.20
gen1 #12	6 cm, upper-right	9.2	7.6	7.6	7.3	0.25
gen2 #106	2 cm, lower-left	6.3	3.0	4.6	2.5	0.40
gen2 #104	3 cm, central	4.9	0.9	1.9	0.9	0.50
gen2 #105	3 cm, upper-left	6.6	0.3	0.3	0.3	0.55
gen3 #6	2 cm, upper-right	3.3	3.0	3.0	3.0	1.00
gen3 #16	3 cm, upper-left	3.0	3.2	3.3	3.3	1.00
gen3 #48	3 cm, lower-right	3.3	3.2	3.0	3.0	1.00

Why use the average and not one scan's best? The last column shows that each scan prefers a different value, anywhere from low to high. A value tuned to one scan can be poor on the next. The average-best value is the safest single setting across the whole set. Some scans (gen1 #7, gen1 #12) stay more than 5 cm from the tumor with both textbook and chosen settings, so no knob value fixes them; that problem lies elsewhere in the pipeline.

5. Limits of this test

- It used 12 carefully mixed scans, not all 2,264 in the dataset. A different mix could shift the result slightly.
- Values were tried in steps of 0.05. Something like 0.32 was not tested separately.
- Scores that differ by less than 1 point out of 100 are treated as practically the same.
- Antenna layout and wave speed were kept as the app already uses them; only the four knobs were tested.

Full numbers for every value are in beamformer_constant_sweep_detailed.pdf and .md in the same results folder.